

INDIAN INSTITUTE OF TECHNOLOGY KANPUR

SPA617 — RADIO ASTRONOMY LABORATORY

Experiment 3: Antenna Characterization

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1 Aim

The aim of this laboratory experiment are:

1. To measure the far-field power radiation pattern of the antenna under test (AUT) over a complete 360° sweep in the E-plane and H-plane co-polarization configurations
2. To measure the corresponding E-plane and H-plane cross-polarization configurations and evaluate the polarization purity and isolation of the antenna system.
3. To determine the half-power beamwidths (HPBW) in both planes (θ_E and θ_H) and estimate the antenna directivity (D) using practical approximations.

2 Apparatus

The following equipment and components were utilized in this experimental setup :

- RF Signal Generator (source of the carrier signal)
- Spectrum analyzer (configured for received power measurements)
- Transmitting Horn/Monopole Antenna and Receiving Dipole Antenna
- High-quality Coaxial Cables, suitable RF adapters (SMA to N-type)

3 Theory

An antenna works as a transducer which converts the EM wave signals to electrical signals and vise versa.

3.1 Distance Decay and the Inverse-Square Law

The power radiated by the antenna is distributed equally around the sphere at a R distance from the source as it propagates. The energy is same but as the distances increases the energy distributed over the surface area is divided more. Consequently, the received power (P_r) is inversely proportional to the square of the distance (R) between the transmitting and receiving antennas.

$$P_r \propto \frac{1}{R^2} \quad \Longrightarrow \quad P_r = \frac{k}{R^2} \tag{1}$$

where k is a system constant. When expressed on a logarithmic decibel scale, the power relationship is linearized against logarithmic distance:

$$P_{r,\mathrm{dB}} \propto 10\log_{10}\left(\frac{k}{R^2}\right) = \text{Constant} - 20\log_{10} R \tag{2}$$

Plotting P_r versus $20\log_{10} R$ yields a straight line with a theoretical slope of -1.0 in free space. Doubling the separation distance ($R \to 2R$) leads to a theoretical drop in power of:

$$\Delta P_r = -20\log_{10}(2) \approx -6.02 \text{ dB} \tag{3}$$

3.2 Radiation Patterns and Polarization

There are different kind of antennas, some radiate isotropically and some radiates in a particular direction which leads to different kind of radiation pattern over the angluar distribution over two orthogonal patterns.

- **E-Plane**: The plane containing the electric-field vector and the propagation vector.
- **H-Plane**: The plane containing the magnetic-field vector and the propagation vector.

Co-polarization is measured when the electric-field vectors of the transmitter and receiver are parallel. **Cross-polarization** is measured by rotating the transmitting antenna by 90°, making its electric field perpendicular to the receiver. On-axis polarization isolation is defined as the decibel difference between co- and cross-polarized peaks at 0°, which indicates the antenna's polarization discrimination.

3.3 Half-Power Beamwidth (HPBW) and Directivity

The half-power beamwidth (θ_E or θ_H) is defined as the angular separation between the two points where the main-lobe received power drops to half of its peak value (representing a $-3\,\text{dB}$ level).

Directivity (D) measures how concentrated the radiation is compared to an isotropic radiator. For practical directional antennas, the beam solid angle (Ω_A) is approximated using the orthogonal half-power beamwidths (expressed in radians):

$$D \approx \frac{4\pi}{\theta_E \theta_H} \tag{4}$$

4 Experiment Setup

The physical system was set up inside the laboratory environment according to the manual's specifications:

1. The transmitting antenna was connected to the RF signal generator, and the receiving antenna was connected to the spectrum analyser. Both antennas were aligned at the same physical height as shown in figure 1.
2. The signal generator was set to transmit a stable continuous-wave signal at 1.4 GHz.
3. To fulfill the far-field criteria, the separation distance was kept at a minimum of 3λ ($\approx 64\,\text{cm}$ at 1.4 GHz) during all angular sweeps. Nearby scattering objects were cleared away to reduce reflections.

5 Observation & Graphs

This section compiles all primary measurement data recorded during the laboratory session.

5.1 Part B: Received Power vs. Distance Data

The received power P_r was measured at a constant operating frequency of 1.4 GHz for distance values R ranging from 261 cm to 674 cm.

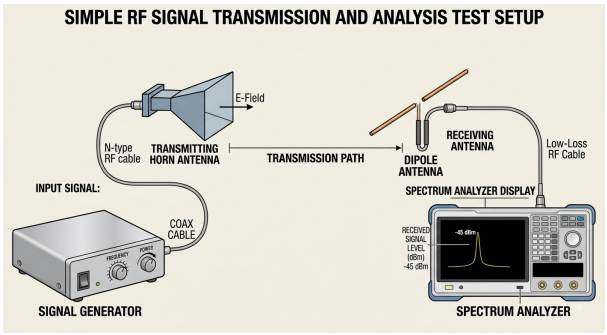


Figure 1: Experiment setup

We got the values of power in Decibels but to plot the power vs R graph we need power in watt form, so we converted it using the formula -

$$P_{\mathrm{W}} = 10^{\frac{P_{\mathrm{dBW}}}{10}}\ \mathrm{W} \quad \Longrightarrow \quad P_{\mu\mathrm{W}} = 10^{\frac{P_{\mathrm{dBW}}}{10}} \times 10^{6}\ \mu\mathrm{W} = 10^{\frac{P_{\mathrm{dBW}}+60}{10}}\ \mu\mathrm{W}$$

After conversion the values are given in table 1 and the the plot between them is shown in the figure 2.
Now we also plotted the power in decibals vs 20log10(R) in figure 3. The values are shown in the table 2 - (the formula was discussed in the theory section.)

Table 1: Part B: Received Power Converted to Microwatts (μW) vs. Separation Distance (cm)

Distance R (mm)	Received Power P_r (μW)	Distance R (mm)	Received Power P_r (μW)
261	1.000	390	0.955
262	0.708	410	1.479
292	1.148	417	1.778
325	1.778	425	0.724
351	1.445	500	0.442
383	0.841	674	0.251

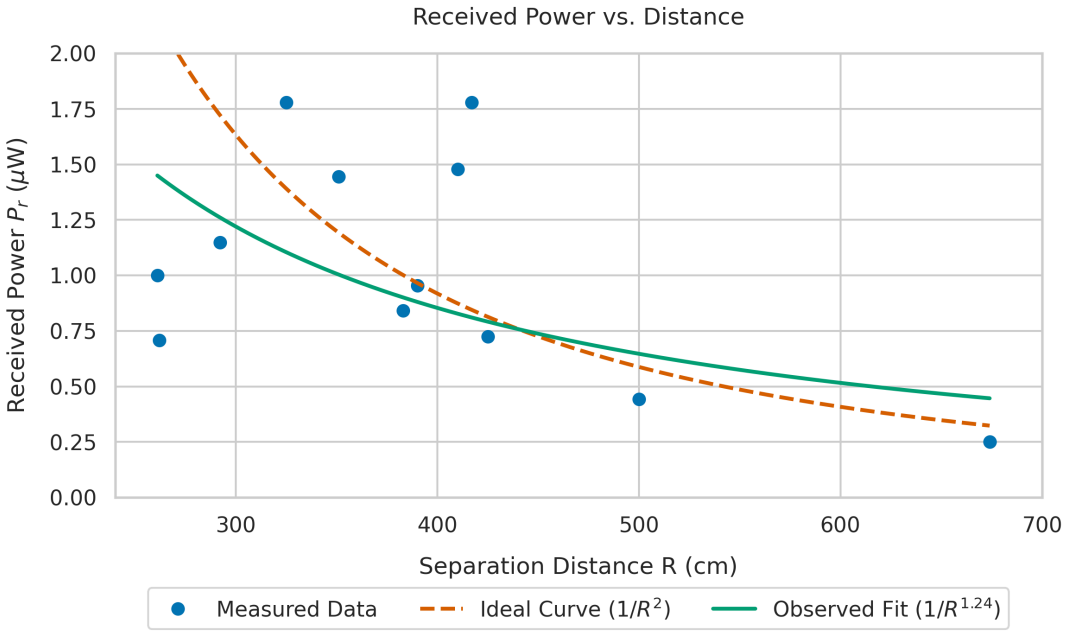


Figure 2: Received Power (P_r) in dbi converted to micro watt to plot with the distance (R) . The linear regression yields a logarithmic decay slope of -0.83.

Table 2: Part B: Received Power (dBm) vs. Separation Distance R (cm)

Distance R (cm)	Received Power P_r (dBm)	Distance R (cm)	Received Power P_r (dBm)
261	-30.00	390	-30.20
262	-31.50	410	-28.30
292	-29.40	417	-27.50
325	-27.50	425	-31.40
351	-28.40	500	-33.55
383	-30.75	674	-36.00

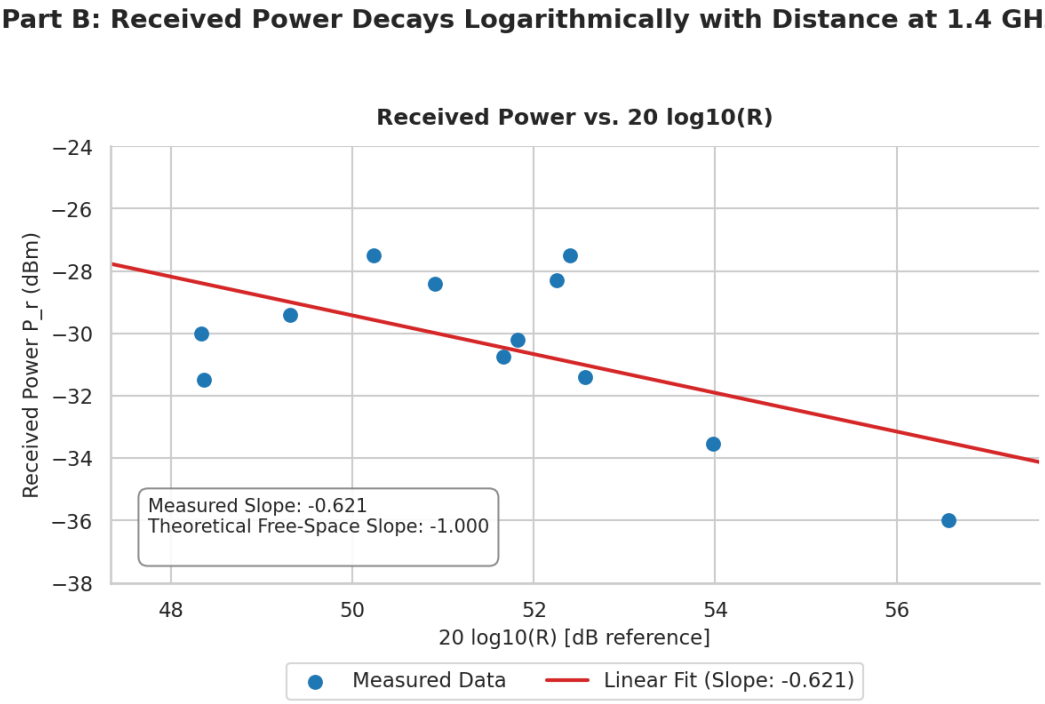


Figure 3: Received Power (P_r) in dbi plotted against the distance (R) in cm . The linear regression yields a logarithmic decay slope of -0.6.

5.2 Part B observation -

- As we can see from the Figure 1 the slope between the power and distance is following the Ideal curve approximately, proving the inverse sqaure distance relation with the power .
- Similiary in the Figure 3 we can see the perfect fit of the curve is a straight line with a slope -0.6 clsoe to -1 which again proves the relation between the logarithmic scale of the power and distance as we discussed in the theory section.

5.3 Part C & D: Radiation Patterns

Table 3 displays the complete co-polarization measurements from 0° to 360°. Table 4 contains the measured cross-polarization values from $-90°$ to 90°.

Table 3: Part C: Co-Polarization E-Plane and H-Plane Angular Readings

Angle (°)	**E-Plane (dBm)**	**H-Plane (dBm)**	**Angle (°)**	**E-Plane (dBm)**	**H-Plane (dBm)**
0	−31.72	-30.16	190	-43.21	-43.63
10	−33.13	-30.56	200	-42.90	-41.15
20	−34.59	-31.47	210	-42.51	-40.30
30	−36.43	-30.99	220	-44.06	-39.87
40	−39.23	-33.72	230	-44.03	-43.67
50	−40.45	-34.98	240	-44.12	-50.05
60	−39.20	-37.33	250	-47.97	-55.20
70	−36.50	-40.06	260	-48.05	-54.90
80	−38.00	-43.46	270	-42.97	-47.20
90	−37.76	-46.28	280	-40.12	-43.63
100	−35.42	-47.04	290	-38.02	-40.76
110	−34.10	-44.02	300	-36.64	-38.60
120	−42.10	-42.63	310	-35.06	-37.42
130	−45.20	-46.59	320	-33.46	-35.07
140	−49.02	-48.98	330	-32.15	-33.62
150	−45.02	-49.70	340	-31.02	-31.02
160	−43.06	-49.23	350	-30.72	-30.19
170	−42.08	-52.80	360	-32.32	-31.72
180	−43.11	-53.00			

Table 4: Part D: Cross-Polarization E-Plane and H-Plane Angular Readings

Angle (°)	**E-Plane (dBm)**	**H-Plane (dBm)**	**Angle (°)**	**E-Plane (dBm)**	**H-Plane (dBm)**
-90	−60.00	-42.27	10	−51.31	-41.19
-80	−55.50	-44.04	20	−48.20	-40.00
-70	−48.40	-46.09	30	−47.30	-39.27
-60	−54.71	-45.24	40	−46.80	-40.44
-50	−52.62	-46.57	50	−47.20	-42.42
-40	−49.17	-49.62	60	−50.50	-43.42
-30	−46.01	-54.07	70	−51.80	-47.13
-20	−49.30	-49.15	80	−49.50	-48.00
-10	−49.40	-43.22	90	−49.80	-48.90
0	−51.50	-43.64			

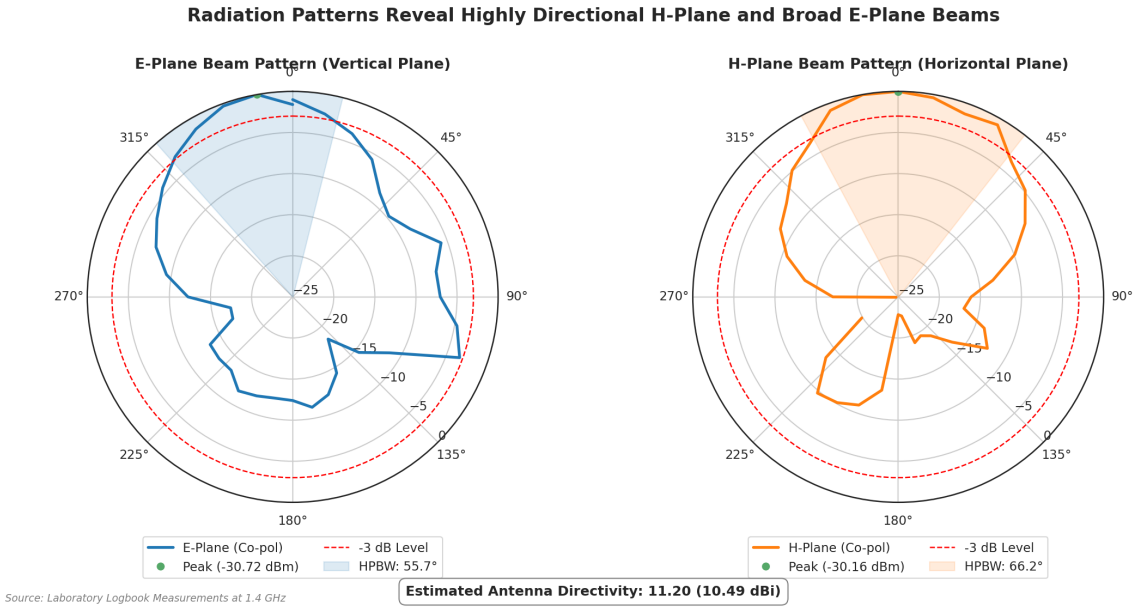


Figure 4: Individual normalized polar radiation patterns displaying E-plane and H-plane co-polarization and cross-polarization beam shapes.

6 Part E: Half-Power Beamwidth and Directivity

6.1 Normalized Beam Pattern Plots

The co-polarisation radiation patterns were normalised to their respective peaks ($E_{\max} = -30.72\,\text{dBm}$, $H_{\max} = -30.72\,\text{dBm}$) and plotted in both polar and rectangular representations.

6.2 HPBW Calculations via Linear Interpolation

Since measurements were recorded at discrete 10° intervals, linear interpolation was used to locate the exact crossing points at the $-3.0\,\text{dB}$ threshold:

$$\theta_{\text{cross}} = \theta_1 + \frac{y_{\text{threshold}} - y_1}{y_2 - y_1} \times (\theta_2 - \theta_1) \tag{5}$$

6.2.1 E-Plane HPBW (θ_E)

The maximum power of the E-plane pattern is $-30.72\,\text{dBm}$ at 350° (normalized peak = 0 dB). The half-power threshold is $y_{\text{threshold}} = -33.72\,\text{dBm}$.

- **Left Crossing (θ_{left}):** Interpolating between 310° ($-35.06\,\text{dBm}$) and 320° ($-33.46\,\text{dBm}$):

$$\theta_{\text{left}} = 310^\circ + \frac{-33.72 - (-35.06)}{-33.46 - (-35.06)} \times (320^\circ - 310^\circ) = \mathbf{318.38}^\circ \tag{6}$$

- **Right Crossing (θ_{right}):** Interpolating between 10° ($-33.13\,\text{dBm}$) and 20° ($-34.59\,\text{dBm}$):

$$\theta_{\text{right}} = 10^\circ + \frac{-33.72 - (-33.13)}{-34.59 - (-33.13)} \times (20^\circ - 10^\circ) = \mathbf{14.04}^\circ \tag{7}$$

The total E-plane half-power beamwidth (θ_E) spanning the North axis is:

$$\theta_E = (360^\circ - 318.38^\circ) + 14.04^\circ = \mathbf{55.67}^\circ \quad (\mathbf{0.9716}\text{ radians}) \tag{8}$$

6.2.2 H-Plane HPBW (θ_H)

By selecting the peak power of the H-plane pattern to be the physical boresight value of $P_{\max} = -30.16\,\text{dBm}$ at 0° (normalized peak = 0 dB), the half-power threshold is calculated as:

$$y_{\text{threshold}} = -30.16\,\text{dBm} - 3.0\,\text{dB} = \mathbf{-33.16}\,\text{dBm} \tag{9}$$

Using linear interpolation on the discrete measurement points, the exact angles where the main lobe crosses this $-33.16\,\text{dBm}$ threshold are determined:

- **Left Crossing (ϕ_{left}):** Interpolating between the discrete points at 330° ($-33.62\,\text{dBm}$) and 340° ($-31.02\,\text{dBm}$):

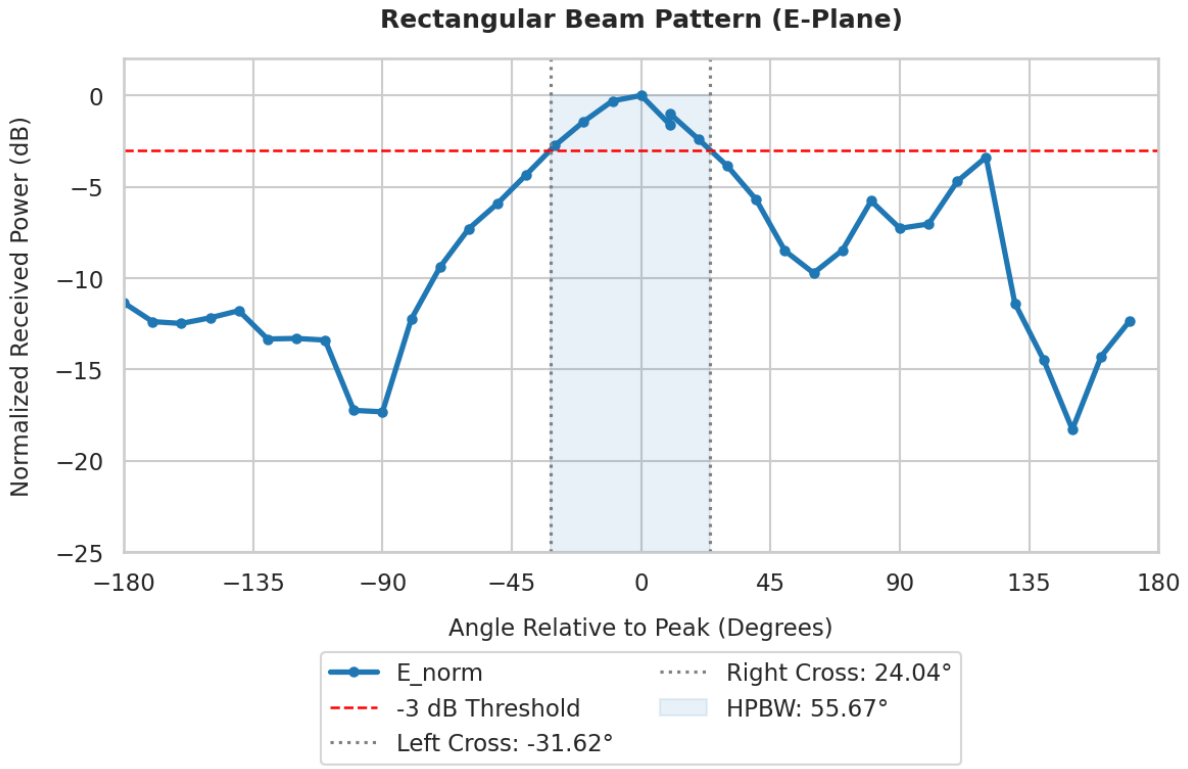
$$\phi_{\text{left}} = 330^\circ + \frac{-33.16 - (-33.62)}{-31.02 - (-33.62)} \times (340^\circ - 330^\circ) = \mathbf{331.77}^\circ \tag{10}$$

- **Right Crossing (ϕ_{right}):** Interpolating between the discrete points at 30° ($-30.99\,\text{dBm}$) and 40° ($-33.72\,\text{dBm}$):

$$\phi_{\text{right}} = 30^\circ + \frac{-33.16 - (-30.99)}{-33.72 - (-30.99)} \times (40^\circ - 30^\circ) = \mathbf{37.95}^\circ \tag{11}$$

The total H-plane half-power beamwidth (θ_H) is the angular span spanning the primary propagation axis:

$$\theta_H = (360^\circ - 331.77^\circ) + 37.95^\circ = 28.23^\circ + 37.95^\circ = \mathbf{66.18}^\circ \quad (\mathbf{1.1551}\text{ radians}) \tag{12}$$



Source: Laboratory Logbook Measurements at 1.4 GHz

Figure 5: Rectangular co-polarisation radiation patterns at 1.4 GHz for the E-plane normalised to their respective main-beam peaks. The horizontal red dashed lines indicate the −3.0 dB half-power threshold. The green indicators mark the exact left and right crossing points calculated via linear interpolation, illustrating the angular span of the half-power beamwidths (HPBW = 55.67° and 66.18°, respectively).

6.3 Antenna Directivity Calculation

Using the recalculated H-plane beamwidth ($\theta_H = 1.1551$ rad) and the previously determined E-plane beamwidth ($\theta_E = 0.9716$ rad), the estimated linear directivity (D) is calculated using the analytical approximation:

$$D \approx \frac{4\pi}{\theta_E \theta_H} = \frac{4\pi}{0.9716 \times 1.1551} = \frac{12.5664}{1.1223} = \mathbf{11.20} \tag{13}$$

Converting this linear value to decibels relative to an isotropic source (dBi):

$$D_{\text{dBi}} = 10 \log_{10}(11.20) = \mathbf{10.49}\ \text{dBi} \tag{14}$$

6.4 Part E Summary Results

After calculating the values we have plotted the normalised power (db) vs Angle for both the E plane and H plane of co-polarisation as shown in figure 5 and 6.

The recalculated directive properties of the horn antenna are summarized in Table 5.

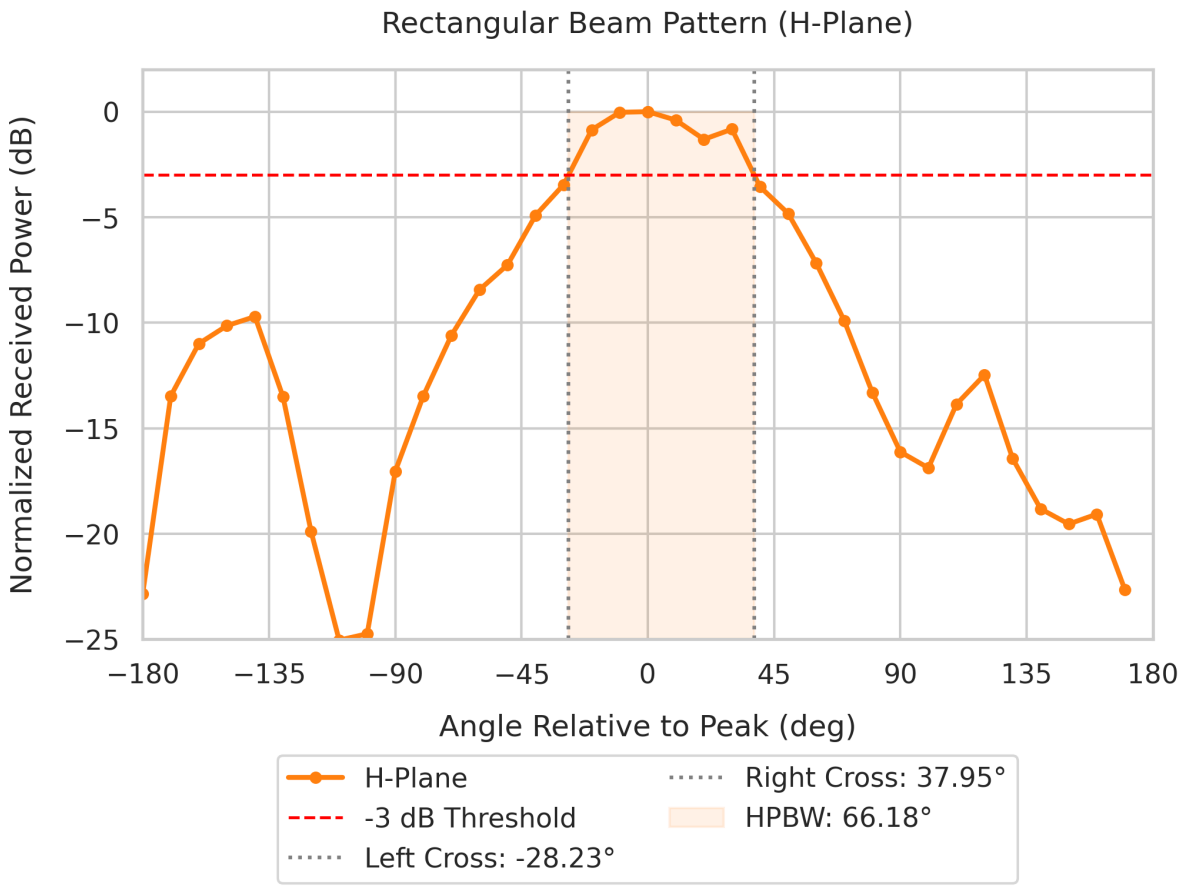


Figure 6: Rectangular co-polarisation radiation patterns at 1.4 GHz for H-plane (right), normalised to their respective main-beam peaks. The horizontal red dashed lines indicate the −3.0 dB half-power threshold. The green indicators mark the exact left and right crossing points calculated via linear interpolation, illustrating the angular span of the half-power beamwidths (HPBW = 55.67° and 66.18°, respectively).

Table 5: Recalculated Antenna Directive Properties (Peak $P_{\max} = -30.16$ dBm)

Parameter	Symbol	Value in Degrees	Value in Radians
E-Plane HPBW	θ_E	55.67°	0.9716
H-Plane HPBW	θ_H	66.18°	1.1551
Directivity (Linear)	D	11.20	
Directivity (Decibels)	D_{dBi}	10.49 dBi	

7 Conclusion

From this experiment we learned about the basic working of antenna and the frequency corresponding the length of the dipol. We also understood the various characteristic og antenna and their relations, similiar to the distance relation with the power of the signal and also how antenna have a beam pattern which can be used to direct our signal in a aprticular direction with minimal loss.

By measuring the half-power beamwidths (HPBW) in both principal planes, we determined the E-plane HPBW (θ_E) to be 55.67° (0.9716 rad) and the H-plane HPBW (θ_H) to be 66.18° (1.1551 rad). These values were used to caluclate the antenna directivity of 11.49 in linear

terms, which translates to 10.49 dBi. This mathematically demonstrates how effectively the horn antenna concentrates its radiated power along its boresight axis compared to an isotropic radiator.

We also observed from the beam pattern which we plotted that the antenna response is not limited to a single direction, but shows significant responses at multiple angles. This is visually represented by the minor side lobes and back lobes mapped in our principal 2D patterns.

7.1 Sources of Experimental Uncertainty

Minor experimental deviations from standard electromagnetic theory are attributed to the following real-world factors:

1. **Coaxial Cable Flexure and Loss:** Manual rotation of the receiving antenna stand through 360° caused continuous twisting of the connected coaxial cables. This mechanical flexure introduces minor dynamic changes in impedance and cable attenuation, explaining the 0.99 dB difference observed between the sweep boundaries at 0° and 360°.

2. **Mechanical Dial Backlash:** The manual rotational stand exhibits a backlash error of approximately $\pm 1°$ to $\pm 2°$. This leads to minor orientation offsets between different measurement runs.

3. **Scattering and Multipath Interference:** Reflections from nearby laboratory walls, metal structural beams, and personnel movements introduced space-dependent constructive and destructive interference, fluctuating the minor lobes and smoothing out null depths in our plots.

8 AI Usage Declaration

The following uses of Artificial Intelligence tools (specifically, LLM assistance) were made during the preparation of this report:

1. **Plot Smoothing and Quality Enhancement:** AI was used to assist in writing and refining Python scripts already written by me to process the experimental measurements. This included implementing techniques for smoothing data curves, adjusting plot layouts, and enhancing the overall visual quality of the distance-dependence and polar beam pattern plots.

2. **LaTeX Formatting and Document Structure:** AI was used as a guide to understand LaTeX formatting, helping to structure the document, set up packages, manage margins, and debug compilation errors to ensure a clean and professional presentation.

3. **Polishing and Refining Written Content:** AI assisted in editing the draft to improve grammatical clarity, sentence flow, and academic tone while preserving the original experimental observations, interpretations, and physical reasoning

4. **Conceptual Clarification:** AI was used as a supplementary tool to clarify theoretical concepts discussed during the laboratory, such as polarization planes (E-plane and H-plane), antenna directivity approximations, and the physical principles of wave propagation in reflective environments.

9 Viva questions

Q1: What is the difference between a horn antenna and a dipole antenna?
Answer:

- **Dipole Antenna:** An omnidirectional (in its azimuth plane), wire-type antenna with low directivity (~ 2.15 dBi) and broad beamwidth.
- **Horn Antenna:** A highly directional, aperture-type antenna formed by flaring a waveguide, providing high gain, narrow beamwidth, and better impedance matching to free space.

Q2: Why is the shape of a horn antenna flared, and does it help in any way or is it random?
Answer: The flared shape is intentional, not random. It focuses the electromagnetic energy into a narrow, highly

Q3: The solid angle of an isotropic radiator is 4π sr. Is the directivity of a horn antenna more or less than that of an isotropic radiator?
Answer: The directivity of a horn antenna is **much greater** than 1 (or 0 dBi, the directivity of an isotropic radiator). An isotropic radiator spreads power uniformly in all directions over a solid angle of 4π sr, whereas a horn antenna concentrates power into a much smaller solid angle Ω_A, yielding high directivity:

$$D = \frac{4\pi}{\Omega_A} \gg 1$$

Q4: How can you determine the operating frequency (or wavelength) using the length of a dipole antenna?
Answer: For a resonant half-wave dipole, the physical length L is approximately half of the wavelength ($\lambda/2$). Using the wave equation $c = f\lambda$:

$$L \approx \frac{\lambda}{2} = \frac{c}{2f} \implies f \approx \frac{c}{2L}$$

where $c \approx 3 \times 10^8$ m/s is the speed of light, and f is the resonant frequency.